

$$x^2 + 2y + y^2 = 0 \Rightarrow x = \frac{-y \pm \sqrt{y^2 - 4y^2}}{2} = \frac{-y \pm \sqrt{-3y^2}}{2} = \frac{-y \pm y\sqrt{-3}}{2}$$

Here $x \notin \mathbb{Z}$.

$\therefore (x-y)=0$ is the only viable option, i.e. $x=y$.

$\therefore$ By Defn. 5, f is one-to-one.

d) $f(n) = \lceil n/2 \rceil$

Solution: $f(1) = \lceil 1/2 \rceil = \lceil 0.5 \rceil = 1$ $f(2) = \lceil 2/2 \rceil = \lceil 1 \rceil = 1$
But $1 \neq 2$. $\therefore$ By Defn. 5, f is not one-to-one.

13) Which functions in Exercise 12 are onto?

a) $f(n) = n-1$

Solution: Let, $f(x) = y$ where $y \in \mathbb{Z}$. $\therefore x-1 = y \Rightarrow x = y+1 \in \mathbb{Z}$.
 $\therefore \forall y \in \mathbb{Z}, \exists x \in \mathbb{Z}$ s.t. $f(x) = y$. $\therefore$ By Defn. 7, f is onto. $\square$

b) $f(n) = n^2 + 1$

Solution: Let, $f(x) = 0$. We know $0 \in \mathbb{Z}$. $x^2 + 1 = 0 \Rightarrow x^2 = -1 \Rightarrow x = \pm \sqrt{-1} \notin \mathbb{Z}$.
 $\therefore 0$ has no mapping x in $\mathbb{Z}$ s.t. $f(x) = 0$.

By Defn. 7, f is not onto. $\square$

c) $f(n) = n^3$

Solution: Let, $f(x) = 2$. We know $2 \in \mathbb{Z}$. $\therefore x^3 = 2 \Rightarrow x^3 - 2 = 0$
 $\Rightarrow x^3 - (\sqrt[3]{2})^3 = 0 \Rightarrow (x - \sqrt[3]{2})(x^2 + x\sqrt[3]{2} + (\sqrt[3]{2})^2) = 0$

$\therefore$ Either $x - \sqrt[3]{2} = 0$ or $x^2 + x\sqrt[3]{2} + (\sqrt[3]{2})^2 = 0$

If $x - \sqrt[3]{2} = 0 \Rightarrow x = \sqrt[3]{2} \notin \mathbb{Z}$.
If $x^2 + x\sqrt[3]{2} + (\sqrt[3]{2})^2 = 0 \Rightarrow x = \frac{-\sqrt[3]{2} \pm \sqrt{(\sqrt[3]{2})^2 - 4(\sqrt[3]{2})^2}}{2} \notin \mathbb{Z}$.

$\therefore$ There does not exist $x \in \mathbb{Z}$ s.t. $f(x) = 2$. By Defn. 7, f is not onto. $\square$

d) $f(n) = \lceil n/2 \rceil$

Solution: Let, $f(x) = y$ where $y \in \mathbb{Z}$. $\therefore \lceil x/2 \rceil = y$

If $x = 2y$, then $\lceil x/2 \rceil = \lceil 2y/2 \rceil = \lceil y \rceil = y$

If $x = 2y-1$, then $\lceil x/2 \rceil = \lceil (2y-1)/2 \rceil = \lceil y-0.5 \rceil = y$

We know that $2y$ and $2y-1 \in \mathbb{Z}$.

By Defn. 7, f is onto. $\square$